



The Technologies and Business Models Behind Global Innovation

Prof. Daniel Zanco

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The Technologies and Business Models Behind Global Innovation

How artificial intelligence, automation, and new revenue models are redesigning competition in global retail — and what this means for your business.

Prof Daniel Zanco

MBAUSP
ESALQ

central do
VAREJO

PROFESSOR **DANIEL ZANCO**

Ambassador Professor MBA USP Esalq and Co-Founder of Central do Varejo.



Why this topic, and why now

- Technology is no longer an IT project and has become a business model decision
- Digital native companies are growing by competing with radically different cost structures
- The speed of adoption is constantly increasing

US\$ 900 bn

global AI market in 2026, on its way to US\$ 4 tn by 2035

US\$ 1 tn

global advertising spending is expected to exceed this mark in 2026

71%

of organizations already use generative AI in at least one area (McKinsey, 2025)

US\$ 2.1 tn

size of global social commerce in 2026, growing 29% annually

Every relevant technology only becomes a competitive advantage when it unlocks a new revenue model.

Our path today...



Technologies

The innovations that are changing the way we operate, decide, and compete



Business models

How these technologies become new ways to generate revenue

HOW TO STAY UPDATED IN THIS DYNAMIC AND COMPLEX SCENARIO ??



Technologies that are redesigning business

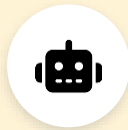
THE TECHNOLOGIES AND BUSINESS MODELS BEHIND GLOBAL INNOVATION

Eight technological fronts to watch

Not every technology on this list has the same stage of maturity — part of strategic insight is knowing how to differentiate hype from real impact.



**Generative AI &
Agents**



**Automation &
Robotics**



**Internet of Things
(IoT)**



Cloud & Big Data



Blockchain & Web3



**Augmented & Virtual
Reality**



5G & Connectivity



Cybersecurity

Generative Artificial Intelligence and Agents

- AI agents perform end-to-end tasks, not just answer specific questions
- The debate has shifted from isolated experimentation to scale, integration, and governance
- In Brazil, maturity is still low — the difference between testing and operationalizing is where the competitive advantage lies

900M

active weekly users of ChatGPT in February 2026

71%

of organizations already use generative AI in at least one function (McKinsey, 2025)

US\$ 900 bn

global AI market in 2026, with a CAGR of ~19% per year until 2035

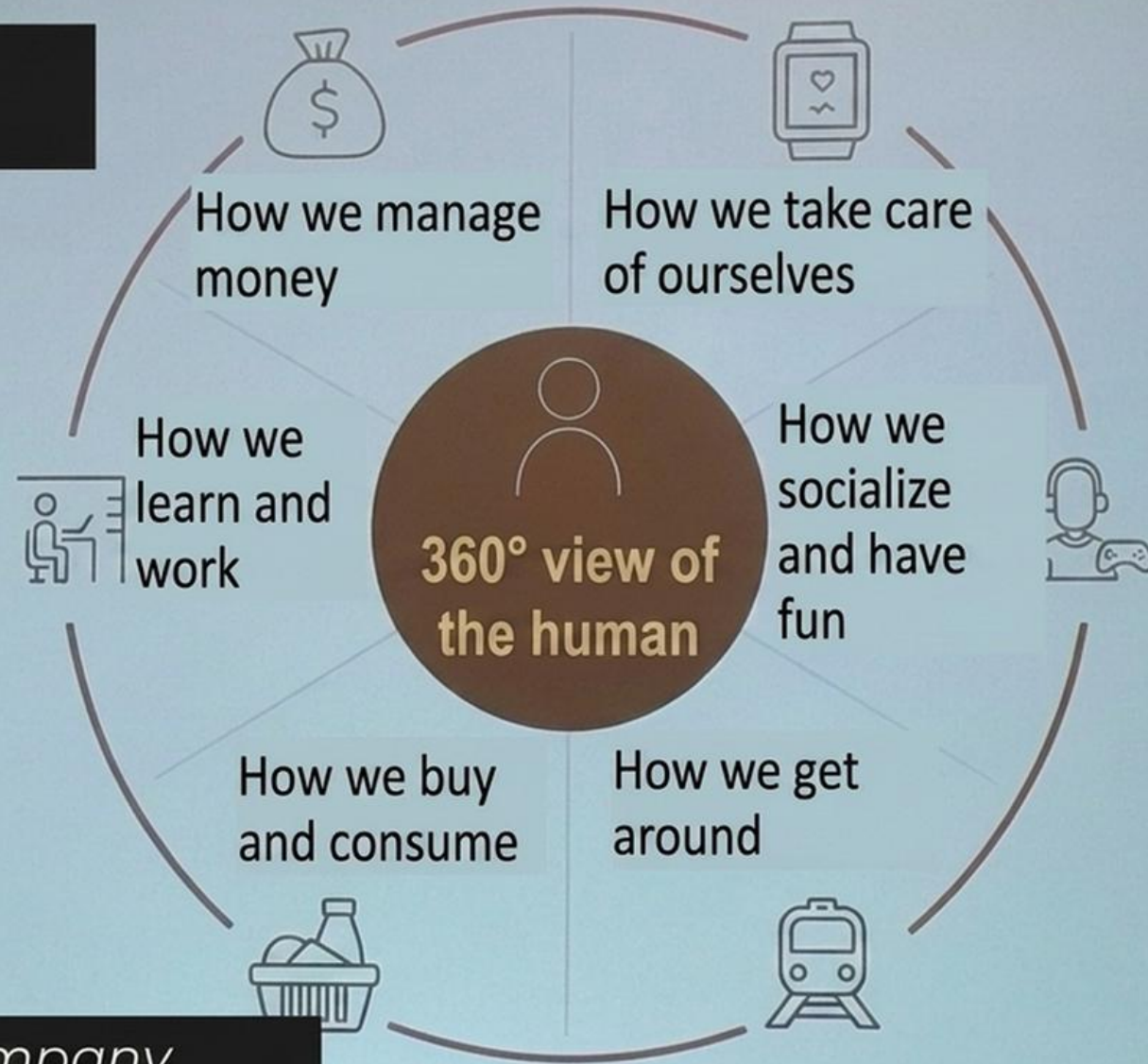
25% → 50%

companies with operational AI agents: projection from 2025 to 2027 (Deloitte)

*Transforming all our
“spheres of life”*

We detect early
signals across different
‘spheres of life’ to
understand the way
people will consume,
exchange or buy
in the future

Consumer Lab - Bain & Company





Generative AI in practice: Brazil and the world

BRAZIL

Magazine Luiza — "Lu"

10+ years

of maturity in data and AI-driven service and recommendation

- One of the first digital personas in Brazilian retail, today powered by generative AI
- International reference as an "influencer" built by a retailer, not an agency
- Proprietary sales data feeds pricing and assortment AI in the marketplace

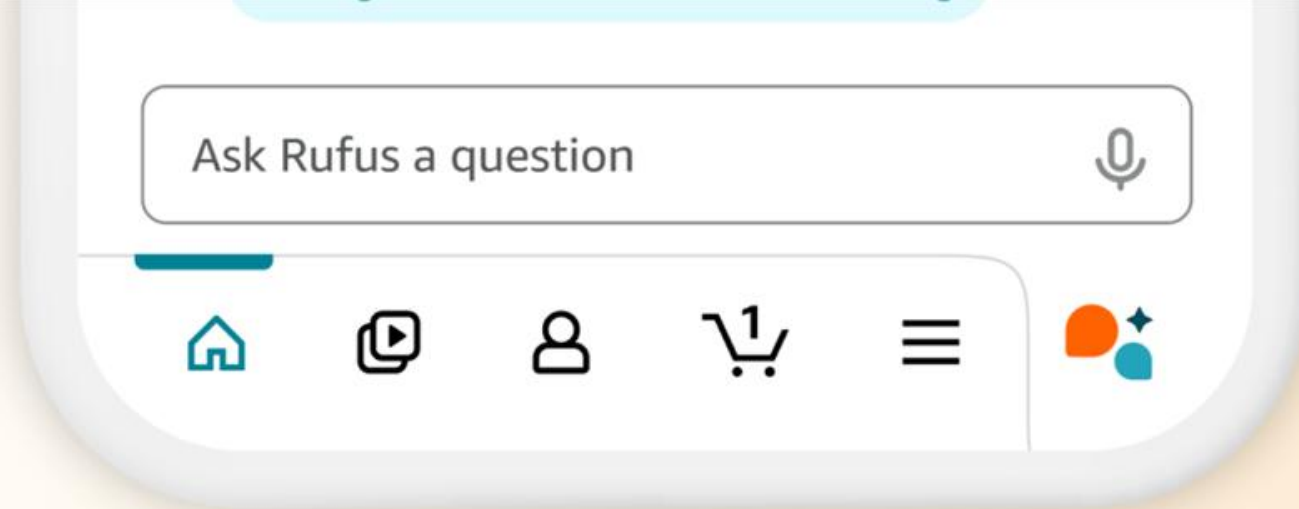
INTERNATIONAL

Amazon Rufus & Microsoft Copilot

80%+

of software vendors must embed generative AI as standard by 2026 (Gartner)

- Rufus, Amazon's shopping assistant, answers questions and compares products within the app
- Copilot is now sold as a standard layer of Microsoft's corporate package, not an extra
- The competitive difference has shifted from "having AI" to "the quality of the data that feeds the AI"



Ask Rufus



Automation and Robotics

US\$ 12.4 bn

global warehouse robotics
market in 2026, growing
28% annually

750 thousand+

robots in operation in
Amazon's distribution
centers today

US\$0.08–0.55

picking cost: from fully
robotic to manual, per unit
separated

25%–40%

operational cost reduction
reported in automated
logistics and manufacturing

- Autonomous mobile robots (AMR) are already the majority of new installations, being cheaper and faster to deploy
- AI has started to act as an "operator": it predicts demand, distributes tasks, and adjusts routes autonomously
- The gain is not just speed — it's a direct and measurable reduction in operational cost per individual order





Internet of Things (IoT)

- RFID, shelf sensors and flow counters are already standard in medium and large-scale operations
- Predictive maintenance prevents silent breakdown from cold storage to agricultural equipment
- Most of the value generated by IoT until 2030 will be captured in B2B applications, not end-consumer

US\$ 1.05 trillion

global IoT market in 2026,
on track for US\$ 5.55 trillion
by 2034

65%

of the value generated by
IoT until 2030 will be
concentrated in the B2B
market

+600%

growth in agribusiness jobs
"augmented" by sensing in
Brazil in 2025

23.1%

projected CAGR for the
global IoT market between
2026 and 2034

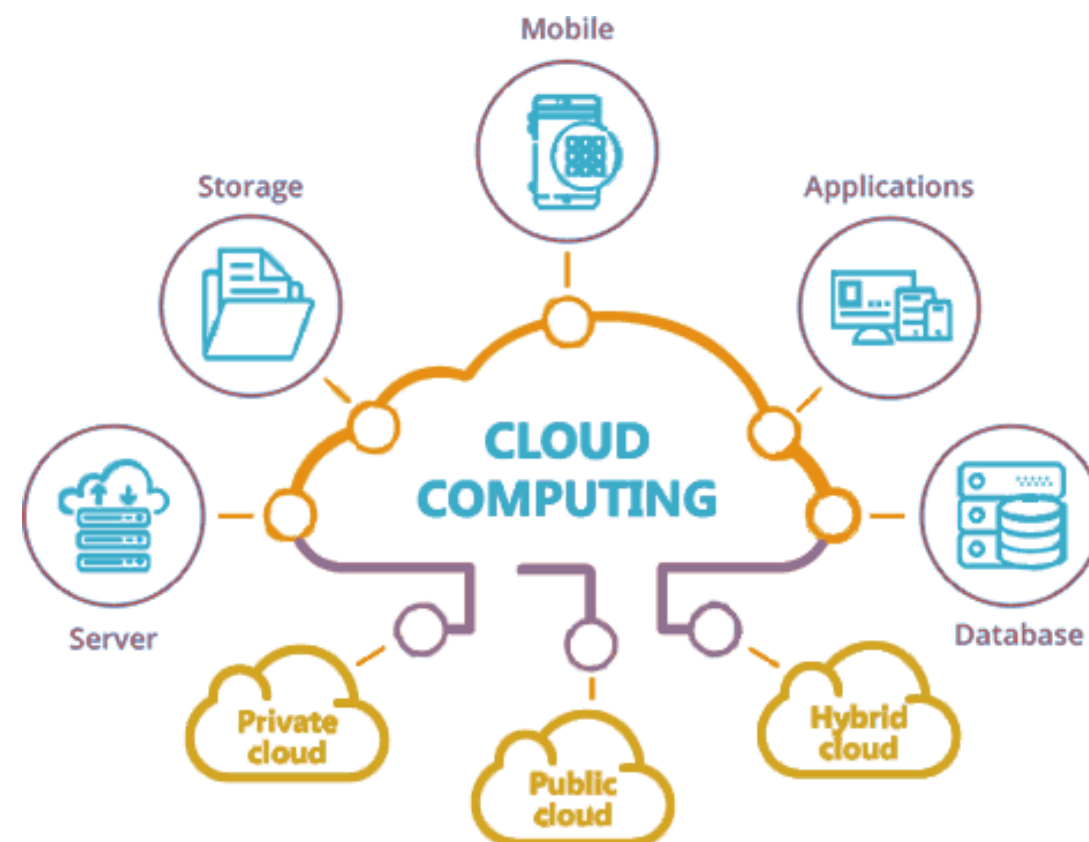


Cloud Computing and Big Data



It's the invisible infrastructure that supports everything else on this list: without cloud and centralized data, there's no AI at scale, no useful IoT, and no 24-hour app.

- Cloud has become the standard model for consuming AI and agents, not just for hosting websites
- Big data is no longer a department and has become an input for every commercial and inventory decision
- Cloud infrastructure for AI is currently the fastest-growing IT category in Brazil



Blockchain and Web3

Beyond the cryptocurrency hype, the most concrete application in retail is trust: proving a product's origin, authenticity, and condition without relying on the word of a single link in the chain.

- End-to-end traceability reduces fraud and increases recall speed when something goes wrong
- Smart contracts automate payments to suppliers when contracted conditions are met
- It is still a niche technology in Brazilian retail — most promising in food, fashion, and luxury

BRAZIL

Food traceability

Food retail chains are testing blockchain to track the origin of meats and produce, giving consumers visibility into the production chain.

INTERNATIONAL

Walmart / IBM Food Trust

Reduced the time to trace the origin of contaminated food from days to seconds — from weeks to seconds, according to the company itself.

Augmented and Virtual Reality

The promise is simple: let the customer "try before buying" even when shopping online — reducing returns and increasing confidence in the decision.

- Virtual fitting rooms and product visualization in real environments already reduce return rates
- It is still a niche technology outside of fashion, beauty, and home decor — but it is growing in other sectors
- The biggest obstacle is no longer technology, but consumer habits and the cost of 3D content

BRAZIL

Beauty and home decor

Virtual makeup try-on and environment simulation are already common features in apps from major beauty and home & construction chains in Brazil.

INTERNATIONAL

IKEA Place and Zara AR

The IKEA Place app projects real-scale furniture inside the customer's home before purchase; Zara has already tested augmented reality fitting rooms in stores.



5G and Connectivity

It is the layer that makes all of this possible simultaneously: robots, sensors, and generative AI depend on fast, low-latency networks to function in real time.

- Low latency enables coordinated industrial automation and cashierless checkout at scale
- 5G coverage is still uneven in Brazil, which delays the adoption of more sophisticated cases outside of capitals
- Stable connectivity is now an operational prerequisite, not a differentiator



INTERNATIONAL

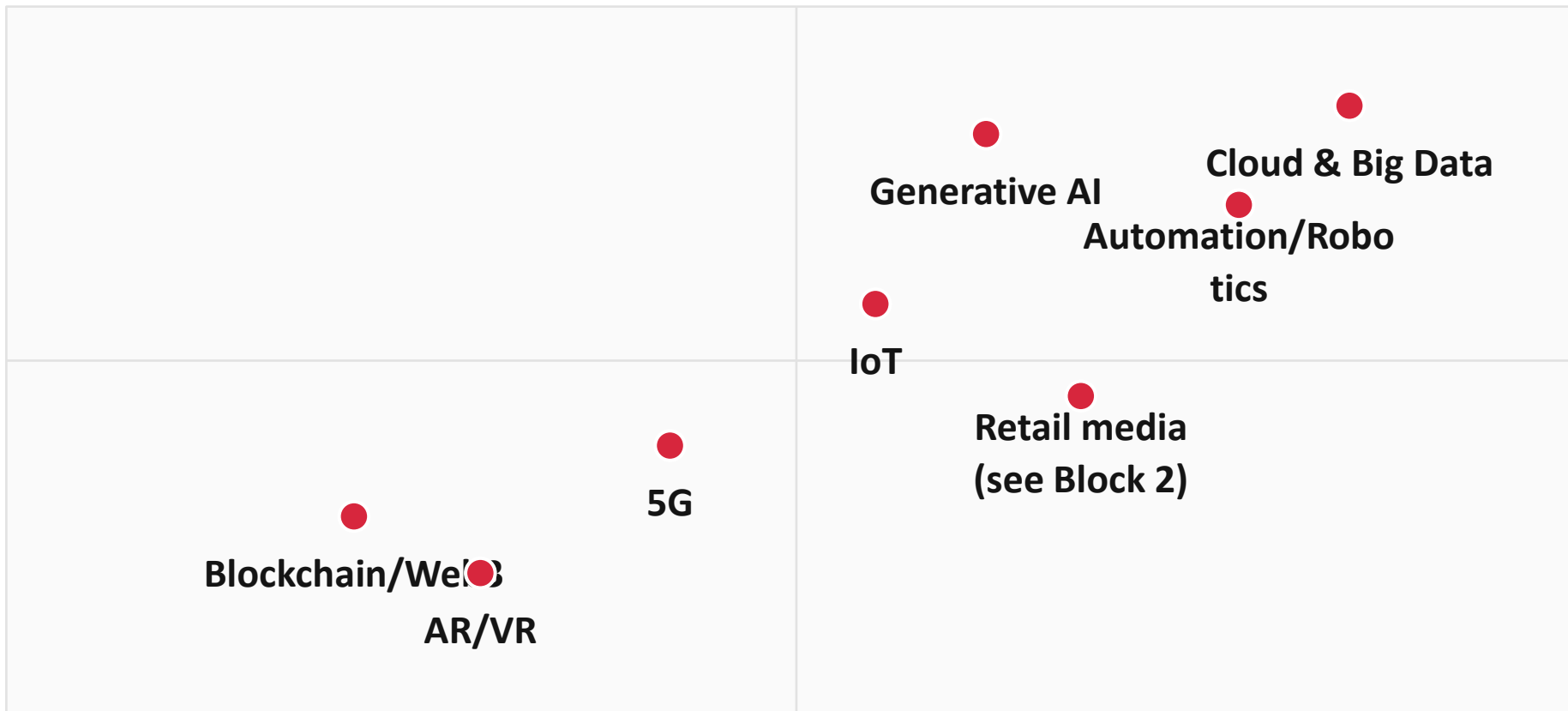
Smart cities and deliveries

Private 5G networks already support drone delivery and port automation in markets such as China, South Korea, and the United States.

Not everything that glitters is a priority right now

Where each technology stands today: how much it already impacts business results

↑ GREATER IMPACT ON RESULTS



MATURITY FOR ADOPTION IN BRAZILIAN RETAIL →



Business models that emerged from these technologies

THE TECHNOLOGIES AND BUSINESS MODELS BEHIND GLOBAL INNOVATION

Eight business models to put on the table

Each one solves a different economic problem — from predictable revenue to idle asset monetization.



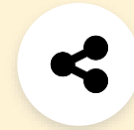
**Subscription
Economy**



**Marketplaces &
Platforms**



**D2C (Direct to
Consumer)**



Sharing Economy



Super Apps



**Live & Social
Commerce**



**Retail Media
Networks**



**Phygital &
Omnichannel**

Subscription Economy

Exchange ownership for access: instead of selling a product once, the company sells the same value every month — and knows today how much it will earn tomorrow.

- Recurring revenue changes how to price, retain, and invest in each customer
- Subscription fatigue is already real: consumers no longer want more subscriptions, they want more value from the ones they already have
- B2B already accounts for more than half of the global subscription market — it's not just streaming

US\$ 738 bn

global subscription economy market in 2026, growing 18.5% annually

78%

of adults maintain at least one active subscription

6,7

is the average number of active subscriptions per person

55%+

of the global subscription market is already B2B, not direct consumption

COMPANIES OPERATING IN BRAZIL WITH THE SUBSCRIPTION MODEL

VIDEO STREAMING 	MUSIC 	READING 	Software 	EDUCATION 	GYMS AND WELLNESS
BEAUTY AND PERSONAL CARE 	FOOD AND DRINKS 	CLOTHES AND ACCESSORIES 	CHILDREN'S 	PET 	Transport
AUTOMOBILES 	CONTENT AND JOURNALISM 	HOME AND DECORATION 	STATIONERY AND ORGANIZATION 	HEALTH 	DIVERSE SERVICES
SUBSCRIPTION BOXES 	HYGIENE AND CLEANING PRODUCTS 	AGRO AND GARDENING 	TECHNOLOGY AND GADGETS 	OTHER EXAMPLES 	AND MUCH MORE! The subscription model is constantly expanding in Brazil!



Marketplaces and Platforms

+56%

Mercado Livre's GMV growth at neutral exchange rates, in the most recent quarter

55% → 85%

Shein's revenue in Brazil from the local seller marketplace, target for the end of 2026

85.5%

of online sales in Brazil concentrated on six major platforms

US\$ 2.1 tn

size of global social commerce in 2026 — marketplace is already a routine purchase





Marketplaces in practice: Brazil and the world

BRAZIL

Mercado Livre and Shein in Brazil

85.5%

of the country's online sales concentrated on the six largest platforms

- Mercado Livre combines marketplace, proprietary logistics, and fintech (Mercado Pago) in a single ecosystem
- Shein migrated from imported operations to a marketplace with local sellers — target of 85% of national revenue
- Small retailers use these platforms as a sales channel without building their own traffic

INTERNATIONAL

Amazon Marketplace

Majority of the volume

of units sold on Amazon already comes from third-party sellers (3P)

- Amazon charges commission, storage fees (FBA), and advertising from third-party sellers
- The marketplace became the largest source of unit growth, with less need for working capital
- The model inspired global replicas, from Mercado Livre to Shopee

D2C — Direct to Consumer

BRAZIL

O Boticario and Vivara

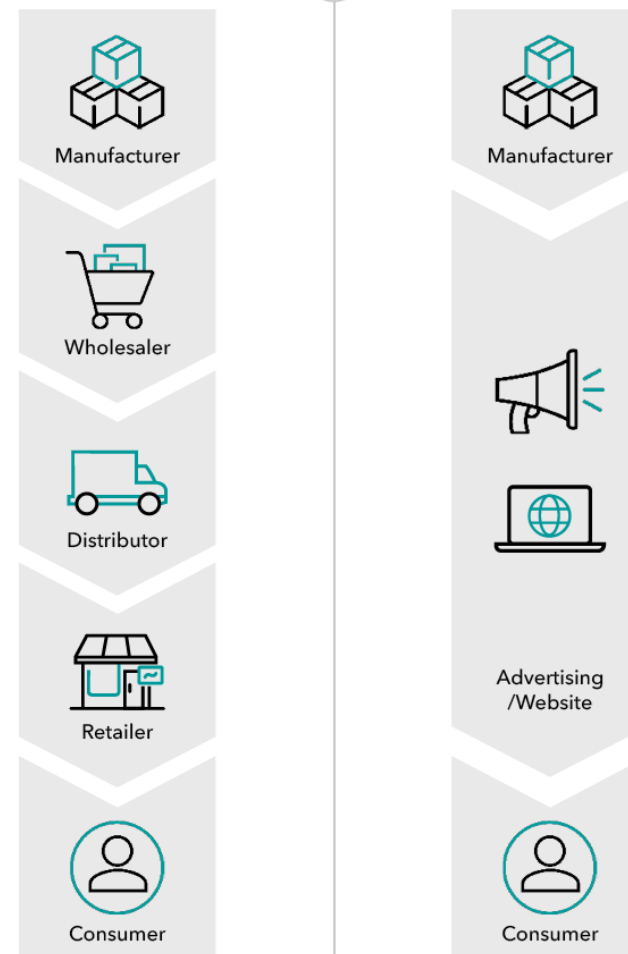
O Boticario combines direct sales by consultants with its own e-commerce, with franchises; Vivara invested in a direct channel and its own app to reduce dependence on shopping centers.

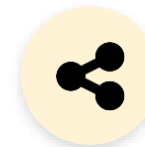
INTERNATIONAL

Nike D2C

Nike reduced its dependence on large multi-brand retailers to sell more directly through its own app and website, seeking margin and customer data.

Business-to-Consumer vs. Direct-to-Consumer





Platforms and Sharing Economy



BRAZIL

Sharing in agriculture and construction

Platforms for renting and sharing agricultural machinery and construction equipment are growing as an alternative to the high cost of buying an asset that sits idle much of the time.

INTERNATIONAL

Uber and Airbnb

Uber monetizes a partner driver's idle car; Airbnb monetizes a host's idle room or property — neither owns the fleet or the properties.

Super Apps

A single app concentrates transport, food, payment, credit, and shopping — the goal is never to give the customer a reason to open another application.

- Each new service within the app increases usage frequency and reduces the acquisition cost of the others
- The transactional data from one service improves the credit, offering, and retention of other services
- It is the most advanced model of cross-monetization within a single customer base

135 million

Nubank global customers —
115 million in Brazil alone,
~61% of the adult
population

+61%

growth in pharmacy orders
within the iFood ecosystem
in one year

255 million

visits to "Nu Shopping", a
marketplace within the
digital bank's app

1 app, N revenues

the super app logic: bank,
marketplace, insurance, and
credit on the same base



Super apps in practice: Brazil and the world

BRAZIL

Nubank and iFood

2nd largest

bank in Brazil by number of customers, surpassing Bradesco, behind only Caixa

- Nubank expanded from credit card to account, investment, insurance, and marketplace in the same app
- iFood stopped being just food delivery and started including pharmacy, grocery, and partnerships with digital banks
- The combination of data between services allows for hyper-personalized credit and discount offers

INTERNATIONAL

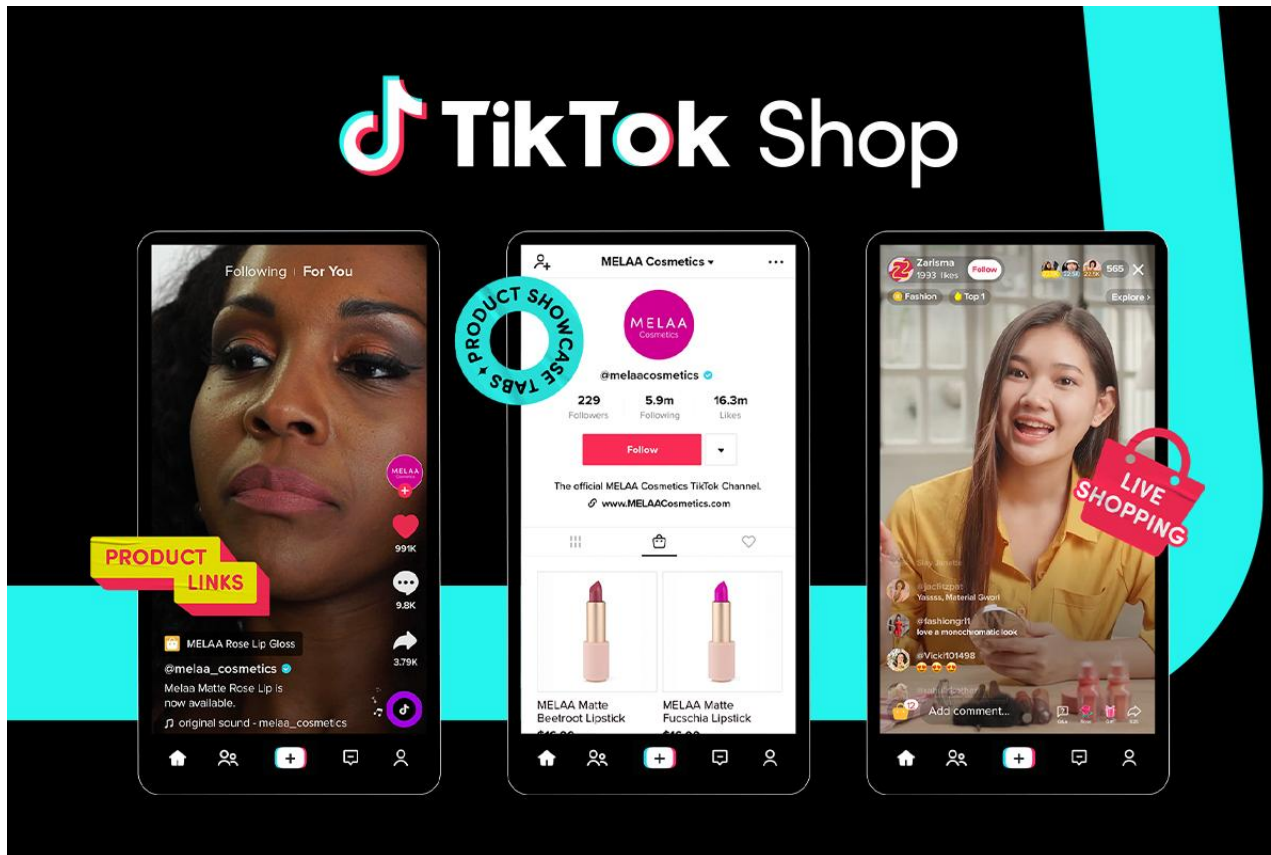
WeChat and Rappi

Chinese model

of global reference: payment, messaging, delivery, and public services in a single app

- WeChat, in China, is the original super app — from chat to electricity bill payment, everything in one place
- Rappi replicates the logic in Latin America, combining delivery, payments, and credit
- The pattern of success is the same: reducing the friction between "needing something" and "solving something"

Live Commerce and Social Commerce



US\$ 2.1 tn
global social commerce
size in 2026, growing 29%
year-on-year

9%–30%
conversion rate in live
commerce, compared to
2%–3% of traditional e-
commerce

~78%
of global live commerce
concentrated in China, led
by Douyin and Taobao Live

US\$ 230 bn
global live commerce
market in 2026 (Grand
View Research)



Live commerce in practice: Brazil and the world

BRAZIL

Shopee Live and brand live streams

Accelerated growth

of the live format within marketplaces and social networks in Brazil

- Shopee Live and promotional live streams from large retailers already combine discounts with live broadcasts
- Brand consultants like Natura use social media for live sales with direct social proof
- Still far from the Chinese scale, but growing faster than traditional e-commerce

INTERNATIONAL

Douyin and TikTok Shop

US\$ 85.3 bn

in sales on Douyin alone (Chinese TikTok) in live streams in the first half of 2026

- In China, 70%–80% of social commerce sales already happen during live streams
- TikTok Shop projects to exceed US\$ 20 billion in sales in the United States in 2026
- The model merges entertainment and checkout: the video doesn't advertise the product, it already sells the product



Retail Media Networks

- Advertising margin is much higher than product margin, using an asset the retailer already has: traffic
- First-party data from the buyer themselves is the main selling asset for this advertising space
- It is already considered the third wave of digital advertising, behind only search and social media

US\$ 197 bn

global retail media market in 2026 (WARC), on track for US\$ 312 bn by 2030

US\$ 69 bn

Amazon's advertising revenue in 2025, 22% annual growth

US\$ 6.4 bn

Walmart Connect revenue in 2025, growing between 37% and 46% annually

25%

estimated share of retail media in global digital advertising spend by 2029

In store



Out of store





Phygital and Omnichannel

- Buying online and picking up in-store (or vice versa) is already a basic consumer expectation, not a differentiator
- Physical store salespeople now operate with the same e-commerce data panel — inventory, history, recommendation
- The physical store gains a new role: less about available inventory, more about experience and convenience

Renner and C&A

Fashion chains integrated app, store inventory, and e-commerce for online purchase with quick pickup, and salespeople accessing the entire network's inventory.

Amazon Fresh and Amazon Go

Amazon's physical stores function as an extension of the digital ecosystem: cashierless, with purchase data feeding the same recommendation engine as the website.

What to watch for starting in 2027



Autonomous AI agents become standard

Deloitte projects a jump from 25% to 50% of companies with operational AI agents between 2025 and 2027 — from assistant to operator.



Retail media continues to eat into the media budget

Projections indicate over 25% of global digital advertising spend will be concentrated in retail media by 2029.



Live commerce is no longer a niche in the West

With conversion rates well above the traditional e-commerce average, the format tends to move out of the experimental phase outside Asia.



The line between technology and model blurs

The decision to invest in AI, automation, or data is no longer an IT agenda item and becomes a central revenue strategy decision.

THE BIG QUESTION :

HOW TO STAY UPDATED IN THIS FAST-PACED WORLD, WITH TECHNOLOGIES AND MODELS THAT CHANGE SO MUCH??

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HOW TO STAY UPDATED IN THIS FAST-PACED WORLD, WITH TECHNOLOGIES AND MODELS THAT CHANGE SO MUCH??

BY SEEKING INNOVATION WHERE IT BEGINS

**DIVE INTO THE MAIN GLOBAL LABORATORY OF INNOVATION
APPLIED TO RETAIL**



CHINA

INNOVATION TOUR

SHANGHAI HANGZHOU SHENZHEN

A MISSION FOR EXECUTIVES WHO WANT TO **DECODE
INNOVATION IN THE CHINESE MARKET.**

NOVEMBER 13 TO 25, 2026 IN CHINA

Schedule* - 10-night package

Day 1	Day 2	Day 3	Day 4	Day 5	DAY 6
Nov. 15	Nov. 16	Nov. 17	Nov. 18	Nov. 19	Nov. 20
Arrival in Shanghai and Hotel Check-in	Hema Freshippo	JD.com Distribution Center	Visit to K11 Art Mall (Japanese luxury retail)	Check-out + Bullet train to Hangzhou	Visit to MyBank (Alibaba digital bank)
	IAPM Mall and NIO House Shopping Experience	Showroom Xiaohong Shu (Little Red Book)	Visit to Iflytek campus	Check-in + Visit to Alibaba Campus + Visit to Wensan Future Center	Technical visit to Cainiao Logistics
Welcome Meeting Dinner	Dinner	Dinner	Dinner	Executive Dinner with Alibaba professional	Dinner

*Preliminary schedule – subject to changes in dates and companies visited.

Agenda* - 13-night package



DAY 7	DAY 8	DAY 9	DAY 10	DAY 11
Nov. 21	Nov. 22	Nov. 23	Nov. 24	Nov. 25
Check-out+ Flight to Shenzhen	Shenzhen Bay Shopping Park Free Morning	Visit to Huawei Campus	Visit to SEG Electronics Market	Return flight to Brazil
Check-in + Free Afternoon	DJI	Visit to Tencent HQ + WeChat Pay Demonstration	Visit to UBTECH Robotics Corp	
Dinner	Dinner	Dinner	central do VAREJO day	

**Preliminary schedule – subject to changes in dates, companies visited.*



Schedule group 1

NRF 2027 | central do VAREJO

Thursday	Friday	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
Jan. 07	Jan. 08	Jan. 09	Jan. 10	Jan. 11	Jan. 12	Jan. 13	Jan. 14	Jan. 15
Boarding for NY	Arrival in New York Free morning	Free day Walking Tour Support for planning visits on the free day	NRF DAY 1 with one-way bus to the event*	NRF DAY 2 with one-way bus to the event*	NRF DAY 3 with one-way bus to the event*	central do VAREJO day Event at Columbia University (all day)	Technical Visits Specialized curation:	Morning free
	Group Check-in at the Hotel	Technical Visits For those attending NYC for the first time				 Central do Varejo Party Exclusive Closing Event	central do VAREJO	
	Welcome NY Opening Event	Outlet:						Departure to Brazil

*Preliminary schedule – subject to changes in dates and events.bus.

*The bus for technical visits will be exclusive to participants attending NRF for the first time.

THANK YOU

<https://www.linkedin.com/in/danielzanco/?locale=pt>

Prof Daniel Zanco

MBAUSP
ESALQ

central do
VAREJO